Project 2 · Machine Learning: From Data to Predictions

Who will subscribe to a term deposit?

Predicting the outcome of a bank’s phone calls before they’re made

Tamim Wahage

The problem

A Portuguese bank calls clients to sell them a term deposit. Most calls end with a no.

What I predict: will this client subscribe? (before the call)

For whom: the marketing team planning the call list

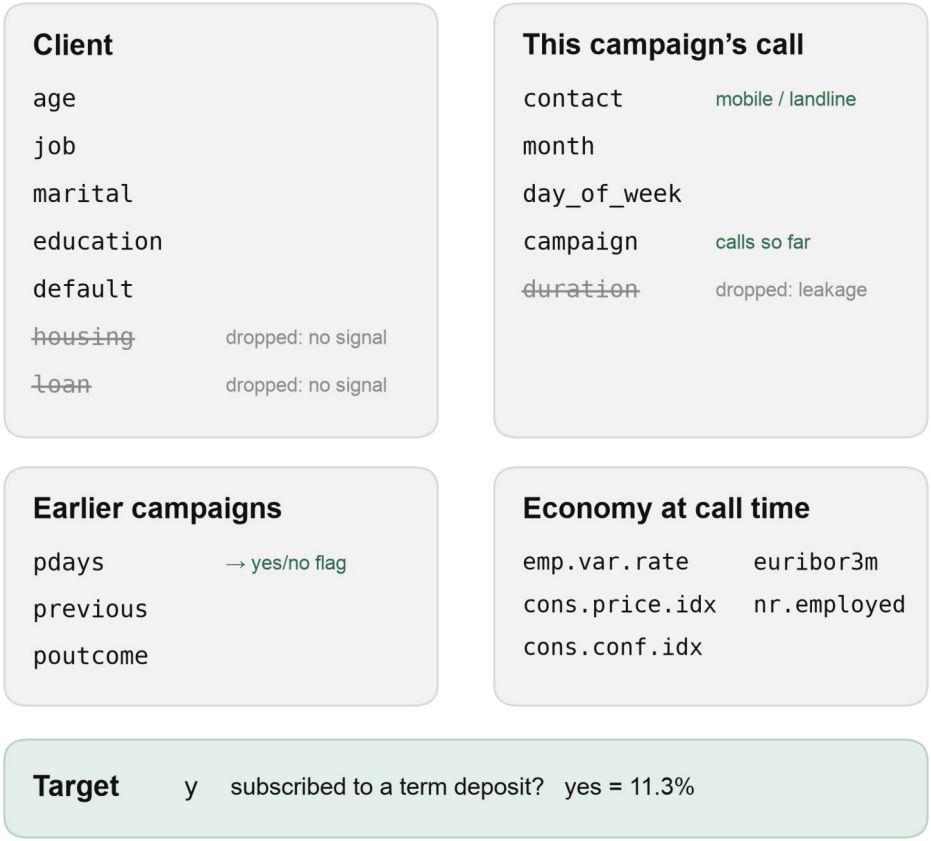
What they do with it: call the likely clients first, skip the ones who’ll almost surely say no

The data

41,188 phone contacts, May 2008 – Nov 2010. One row = one call to one client: client info, earlier contacts, the economy at that time.

The traps I found

- **duration** (call length) is only known after the call → dropped (leakage)
- missing values hide as **“unknown”** and **pdays = 999** → kept as a category / replaced by a flag
- 12 exact duplicates → dropped
- the file is sorted by date and the economy columns act like a hidden date



Source: Moro, Rita & Cortez (2014), Bank Marketing, UCI Machine Learning Repository, doi.org/10.24432/C5K306. Licence: CC BY 4.0.

The metric and the baseline

Only **11.3%** say yes. Saying “no” to everyone is **88.7% accurate** and finds nobody.

A missed yes (lost sale) is worse than a wasted call.

So: **F1**, which is only high if the model finds the yes clients *and* is right when it calls.

Baseline: call everyone

0.203

F1 to beat (finds every subscriber, but 89% of the calls are wasted)

What I tried

4 model families × 6 ways to handle the 11% yes (none, class weights, random oversampling, SMOTE, random undersampling, cluster centroids), same 5 folds, each at its best threshold. Best of each:

Model	Imbalance handling	F1 (mean ± spread)	Recall
Gradient boosting	class weights	0.499 ± 0.014	57%
Logistic regression	random oversampling	0.485 ± 0.012	57%
Random forest	random undersampling	0.466 ± 0.014	50%
Decision tree	SMOTE	0.338 ± 0.014	37%
Baseline: call everyone		0.203	100%

The model family matters, the imbalance method hardly does: once the threshold is moved, every method except cluster centroids lands within the fold spread. Tuning didn’t beat the default settings.

Imbalance and the threshold

Gradient boosting, cross-validated F1 for each way to handle the 11% yes:

Method	at threshold 0.5	at its best threshold
No handling	0.365	0.497
Class weights	0.475	0.499
Random oversampling	0.469	0.497
SMOTE	0.426	0.494
Random undersampling	0.440	0.490
Cluster centroids	0.203	0.209

Don’t leave the threshold at 0.5. F1 is best at 0.65. A missed yes costs more than a wasted call, so I went one step lower to **0.60**: it finds more subscribers, and F1 stays just as good.

The final model on the test set

Test set (used once)

0.519

F1

Cross-validation

0.496

± 0.012 across folds

Baseline (call everyone)

0.203

F1

With **15% of the calls** the model reaches **61% of the subscribers**. Calling everyone: 100% of the calls, and 89% of them are wasted.

How: the test set has 8,236 clients, 928 of them subscribers. The model calls 1,248 of them (15%), and 565 of those subscribe: 565 / 928 = 61%. Calling everyone, only 11.3% say yes, so 89% of the calls are wasted.

Where it goes wrong

The model mostly recognises **the time period**, not the client. Test set, grouped by the economy at the time of the call:

Period (nr.employed)	Clients	Said yes	Called	Found
Worse economy (≤ 5076)	975	46%	96%	98%
Middle (5099)	1,695	14%	17%	50%
Better economy (5191–5228)	5,564	4.5%	0.4%	4%

Also weak: landline clients (finds 33%) and clients never contacted before (finds 48%).

What drives it

Permutation importance (drop in test F1)

Column	Drop
nr.employed (economy)	0.23
cons.conf.idx (economy)	0.05
poutcome (last campaign)	0.02
euribor3m (economy)	0.02
month	0.02

For the bank

When a client is called matters more than **who** they are: in bad economic times many more people took a deposit.

Within a period: past success, a mobile number, being young or retired.

Conclusions and limitations

Future check: trained on older calls, tested on newer ones → F1 **0.398**, worse than calling everyone in that period (0.468).

Would I use it?

- Not as it is for a new campaign
- Use the patterns now: always call past successes, mobile numbers, students and retired
- Don’t call the same client again and again

With another week

- Test on later campaigns instead of a random split
- Keep the economy, but in a form that carries over to new periods (e.g. trends instead of exact values)
- Real call costs and deposit value from the bank
- Retrain on recent campaigns

Live demo

The saved model (pipeline + threshold 0.60) is loaded and asked about one client, a 67-year-old retired person with a mobile number:

Same client, called…	P(yes)	Decision
in a bad economic period (Oct, low rates)	0.81	call
in a good economic period (Jul, high rates)	0.37	skip
good period, but last campaign was a success	0.69	call

notebooks/03_evaluation_and_interpretation.ipynb · section 5

Thank you

Who will subscribe to a term deposit?

Tamim Wahage · Questions welcome